Niraj Jaishwal

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SUMMARY

Data Science and dual-degree CS/Applied Mathematics graduate with hands-on experience building end-to-end web applications, automated financial pipelines, and machine learning solutions. Proficient in developing RESTful APIs with Python/Flask and interactive frontends, with a strong foundation in mathematical rigor for financial platform development.

EDUCATION

University of Akron – Honors College | Akron, OH — May 2026

B.S. in Computer Science | B.S. in Applied Mathematics (Data Science Focus) | Minor in Statistics

GPA: 3.86/4.0 | President's List: Spring 2022, Fall 2024–Fall 2025 | Dean's List: Fall 2022–Spring 2024

SKILLS

Languages Python, JavaScript, C++, SQL, R

Frameworks/Tools Flask, RESTful APIs, Git, VS Code, Azure

Data & Finance PostgreSQL/SQL, Pandas, ETL Pipeline, SAS, SPSS, Mathematical Modeling

EXPERIENCE

Entrepreneurial Analyst Intern — June 2024 – May 2026

University of Akron Research Foundation | Akron, OH

- Built and maintained a longitudinal dataset of 1,000+ NSF I-Corps teams (2013–present), performing end-to-end data cleaning, structuring, and statistical analysis to track grants, assess program outcomes, and support NSF reporting.
- Automated reporting and operational workflows (certificate generation, data management, program reports) using Python scripts, reducing manual effort by 60% and improving data pipeline reliability.
- Facilitated $100,000+ in Spark Fund awards under Ohio Third Frontier Program by synthesizing technical feasibility findings, prior-art searches (USPTO/Google Patents), and BCC Research market data to quantify IP novelty and commercial potential.
- Analyzed discovery interviews and qualitative findings to validate product-market fit for early-stage university technologies in Clean Energy and Applied Materials, translating complex technical findings into actionable insights for selection committees.

Student Technical Support Analyst — April 2024 – May 2026

University of Akron — Information Technology Services | Akron, OH

- Delivered data-driven technical support to 10,000+ students, faculty, and staff by resolving hardware, software, account, and network issues via Microsoft Azure and Microsoft Intune.
- Maintained 200+ laptops using a ticketing system and routine update cycles, reducing mean resolution time by 30% through process optimization and cross-functional IT collaboration.
- Collaborated with cross-functional IT teams to redesign support workflows, resolving recurring technical issue efficiently.

PROJECTS

Stock Price Direction Predictor using LSTM (Deep Learning) | TensorFlow/Keras, Flask, Python

- Developed a binary classification deep learning model using Long Short-Term Memory (LSTM) architectures in TensorFlow/Keras to predict short-term price direction.
- Engineered ETL pipeline to extract raw data via yfinance API, synthesizing six technical indicators (MACD, RSI, ATR) and implemented MinMax Scaling with a time-aware 80/20 split to prevent data leakage.
- Trained a 3-layer LSTM with Adam optimizer and Binary Cross-Entropy loss, achieving 60% validation and 55% test accuracy.
- Evaluated directional reliability, reaching 0.74 recall for downward trends, and deployed the model as a production-ready Flask application with Plotly dashboards on Render.

IMDB Sentiment Analysis using Pretrained BERT | Python, BERT

- Fine-tuned a pretrained BERT transformer model for binary sentiment classification on the IMDB dataset (50,000 reviews via Hugging Face), handling tokenization, padding, and large-scale text preprocessing.
- Built end-to-end training and evaluation pipelines in Python using Hugging Face Transformers; achieved 93.7% validation accuracy across 3 epochs, demonstrating applied transformer and LLM-adjacent model development.

Telecommunication Customer Churn Prediction | Python, R, SPSS, Flask, JavaScript

- Engineered an end-to-end ML pipeline on 7,000+ telecom records, performing feature engineering and EDA (Pandas, Seaborn); compared CHAID, CART, and logistic regression models, achieving 0.83 ROC-AUC with 0.19 misclassification risk.
- Identified contract type and tenure as key churn drivers through CHAID and CART decision tree split analysis; optimized model sensitivity to 62% while maintaining 89%+ specificity.
- Deployed a real-time prediction engine via Flask RESTful API integrated with a JavaScript frontend, enabling interactive stakeholder use and demonstrating end-to-end ML solution.

CERTIFICATIONS & ACTIVITIES

- Generative AI, Coursera
- IBM Machine Learning with Python, Coursera
- Dell TechDirect Certification, Dell
- Treasurer of Everest Student Nepalese Organization
- Mathematics Tutor
- Williams Honors College Peer Mentor